

ITERATED FILTERING

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Can you check and tell me whether this is correct. If so, then I think we have a much simpler proof of Theorem 1.

1. A SIMPLIFIED VERSION OF ITERATED FILTERING

Let $\{\bar{f}_\theta(x, y), \theta \in \mathbb{R}^{d_\theta}\}$ be a statistical model where $\bar{f}_\theta(x, y)$ is a density wrt some generic σ -finite product measure that we denote by $dx dy$ on some measurable space $\mathsf{X} \times \mathsf{Y}$. Suppose that we observe $Y = y$, where $(X, Y) \sim \bar{f}_{\theta_\star}$, for some value $\theta_\star \in \mathbb{R}^{d_\theta}$. We are interested in estimating $\theta_\star$. The observed-log-likelihood is $\ell(\theta|y) \stackrel{\text{def}}{=} \log \int \bar{f}_\theta(x, y) dx$, and the mle of $\theta_\star$ is defined as $\hat{\theta}(y) \stackrel{\text{def}}{=} \text{Argmax}_{\theta \in \Theta} \ell(\theta|y)$. We fix the data to some arbitrary value $y \in \mathsf{Y}$ and consider the problem of computing $\hat{\theta}$. For notational simplicity, we will omit the dependence in y and simply write $\ell(\theta)$ instead of $\ell(\theta|y)$. The gradient and Hessian matrix of ℓ are denoted $\nabla \ell$ and $\nabla^2 \ell$ respectively. One standard approach to computing $\hat{\theta}$ follows from the basic fact that

$$\nabla \ell(\theta) = \int \{\nabla_\theta \log \bar{f}_\theta(x, y)\} \pi_\theta(x|y) dx, \quad (1)$$

where $\pi_\theta(\cdot|y)$ is the conditional density of x given y in the joint $\bar{f}_\theta(x, y)$.

This identity offers the possibility to implement various stochastic approximation algorithms to finding $\hat{\theta}$. But it is not always feasible to compute $\nabla_\theta \log \bar{f}_\theta(x, y)$. The idea behind iterated filtering makes it possible to approximate the derivative of ℓ without having to compute $\nabla_\theta \log \bar{f}_\theta(x, y)$.

Let κ be a mean-zero density (wrt the Lebesgue measure) with a convex compact support. For $\theta \in \mathbb{R}^{d_\theta}$ and $\tau > 0$, let $\kappa_{\theta, \tau}$ be the density of $\theta + \tau Z$, where $Z \sim \kappa$. That is, $\kappa_{\theta, \tau}(u) = \tau^{-1} \kappa(\tau^{-1}(u - \theta))$. We introduce the random object $(\check{X}, \check{Y}, \check{\Theta})$ with density $\bar{f}_\vartheta(x, y) \kappa_{\theta, \tau}(\vartheta)$ and denote $\check{\mathbb{E}}_{\theta, \tau}(\cdot)$ its expectation operator. We impose the following assumptions. We use $\|\cdot\|$ to denote both the Euclidean norm on $\mathbb{R}^{d_\theta}$ and some convenient matrix norm.

A1 $\nabla \ell$ and $\nabla^2 \ell$ are well defined and for any compact subset K of $\mathbb{R}^{d_\theta}$,

$$\sup_{\vartheta \in K} \|\nabla \ell(\vartheta)\| < \infty, \quad \text{and} \quad \sup_{\vartheta \in K} \|\nabla^2 \ell(\vartheta)\| < \infty. \quad (2)$$

A1 implies that $\nabla \ell$ is Lipschitz on compact sets, and we will use this fact below.

Theorem 1.1. *Assume A1-A2. Let $h : \mathbb{R}^p \rightarrow \mathbb{R}^m$ be a measurable function for which there exist $\alpha \geq 1$ and $c \in (0, \infty)$ such that*

$$|h(u)| \leq c|u|^\alpha, \quad u \in K_0.$$

Then for any compact subset K of $\mathbb{R}^p$, we can find a finite constant C such that for all τ small enough

$$\sup_{\theta \in K} \left| \mathbb{E}_{\theta, \tau} (h(\check{\Theta} - \theta) | \check{Y} = y) - \int h(\theta + \tau \vartheta) \kappa(\vartheta) d\vartheta - \tau \left(\int \{h(\theta + \tau \vartheta) \vartheta'\} \kappa(\vartheta) d\vartheta \right) \nabla \ell(\theta) \right| \leq C \tau^{2+\alpha}. \quad (3)$$

Proof. Let $\pi_{\check{Y}|\check{\Theta}}$ denotes the conditional density of $\check{Y}$ given $\check{\Theta}$. Then $\pi_{\check{Y}|\check{\Theta}}(y|\vartheta) = \int \bar{f}_\vartheta(x, y) dx = \exp(\ell(\vartheta|y))$. Then we have:

$$\mathbb{E}_\theta (h(\check{\Theta}) | \check{Y} = y) = \frac{\int h(\vartheta) \pi_{\check{Y}|\check{\Theta}}(y|\vartheta) \kappa_{\theta, \tau}(\vartheta) d\vartheta}{\int \pi_{\check{Y}|\check{\Theta}}(y|\vartheta) \kappa_{\theta, \tau}(\vartheta) d\vartheta} = \frac{\int h(\vartheta) \exp(\ell(\vartheta) - \ell(\theta)) \kappa_{\theta, \tau}(\vartheta) d\vartheta}{\int \exp(\ell(\vartheta) - \ell(\theta)) \kappa_{\theta, \tau}(\vartheta) d\vartheta}.$$

We start with the denominator. By Taylor expansion, $\ell(\vartheta) - \ell(\theta) = (\vartheta - \theta)' \int_0^1 \nabla \ell(\theta_t) dt = O(\|\vartheta - \theta\|)$, where $\theta_t = \theta + t(\vartheta - \theta)$. Again, by Taylor expansion $e^x = 1 + x + \left(\int_0^1 (1-t) e^{tx} dt \right) x^2$, and it follows that

$$\begin{aligned} & \int \exp(\ell(\vartheta) - \ell(\theta)) \kappa_{\theta, \tau}(\vartheta) d\vartheta \\ &= 1 + \int (\vartheta - \theta)' \int_0^1 \nabla \ell(\theta_t) dt \kappa_{\theta, \tau}(\vartheta) d\vartheta + O \left(\int \|\vartheta - \theta\|^2 \kappa_{\theta, \tau}(\vartheta) d\vartheta \right), \\ &= 1 + \int (\vartheta - \theta)' \nabla \ell(\theta) \kappa_{\theta, \tau}(\vartheta) d\vartheta + \int (\vartheta - \theta)' \int_0^1 \{\nabla \ell(\theta_t) - \nabla \ell(\theta)\} dt \kappa_{\theta, \tau}(\vartheta) d\vartheta \\ &\quad + O \left(\int \|\vartheta - \theta\|^2 \kappa_{\theta, \tau}(\vartheta) d\vartheta \right) = 1 + O(\tau^2). \end{aligned}$$

The last equality uses the facts that $\int (\vartheta - \theta) \kappa_{\theta, \tau}(\vartheta) d\vartheta = 0$ and that $\nabla \ell(\theta_t) - \nabla \ell(\theta) = O(\|\vartheta - \theta\|)$. For the numerator, we notice that $\ell(\vartheta) - \ell(\theta) = O(\|\vartheta - \theta\|)$ and by expanding the function e^x to the order 3, we get:

$$\begin{aligned} \int h(\vartheta) \exp(\ell(\vartheta) - \ell(\theta)) \kappa_{\theta, \tau}(\vartheta) d\vartheta &= \int h(\vartheta) \kappa_{\theta, \tau}(\vartheta) d\vartheta + \int h(\vartheta) (\ell(\vartheta) - \ell(\theta)) \kappa_{\theta, \tau}(\vartheta) d\vartheta \\ &\quad + \int h(\vartheta) (\ell(\vartheta) - \ell(\theta))^2 \kappa_{\theta, \tau}(\vartheta) d\vartheta + O(\tau^{\alpha+3}). \end{aligned}$$

We consider each of the terms:

$$\int h(\vartheta) \kappa_{\theta, \tau}(\vartheta) d\vartheta = \int h(\theta + \tau \vartheta) \kappa(\vartheta) d\vartheta,$$

$$\begin{aligned}
\int h(\vartheta) (\ell(\vartheta) - \ell(\theta)) \kappa_{\theta, \tau}(\vartheta) d\vartheta &= \tau \left(\int h(\theta + \tau\vartheta) \vartheta' \kappa(\vartheta) d\vartheta \right) \nabla \ell(\theta) \\
&\quad + \tau^2 \int h(\theta + \tau\vartheta) \left\{ \int_0^1 (1-t) \vartheta' \nabla^2 \ell(\theta_t) \vartheta dt \right\} \kappa(\vartheta) d\vartheta \\
&= \tau \left(\int h(\theta + \tau\vartheta) \vartheta' \kappa(\vartheta) d\vartheta \right) \nabla \ell(\theta) + O(\tau^{2+\alpha}),
\end{aligned}$$

where $\theta_t = \theta + t\vartheta$. For the last term,

$$\int h(\vartheta) (\ell(\vartheta) - \ell(\theta))^2 \kappa_{\theta, \tau}(\vartheta) d\vartheta = \tau^2 \int h(\theta + \tau\vartheta) \left\{ \vartheta' \nabla \ell(\vartheta) \right\}^2 \kappa(\vartheta) d\vartheta + O(\tau^{4+\alpha}) = O(\tau^{2+\alpha}).$$

The results follows. $\square$

The natural choice is to take $h(\vartheta) = \vartheta - \theta$ which leads to

$$\tau^{-2} \Sigma^{-1} \left\{ \check{\mathbb{E}}_{\theta, \tau} (\check{\Theta} | \check{Y} = y) - \theta \right\} = \nabla \ell(\theta) + O(\tau),$$

where Σ is the covariance matrix of κ assumed positive definite.

2. APPLICATION TO STATE SPACE MODELS

For $\theta_{0:n} \in \{\mathbb{R}^{d_\theta}\}^{n+1}$, let $\{X_0, (X_k, Y_k), 1 \leq k \leq n\}$ be a Markovian state space process:

- X_0 has density $\mu_{\theta_0}(x_0)$,
- the conditional density of (X_k, Y_k) given $\sigma\{X_0, X_i, Y_i, 1 \leq i \leq k-1\}$ and $X_{k-1} = x_{k-1}$ is $\bar{f}_{\theta_k}(x, y | x_{k-1})$, for densities $\mu_{\theta_0}, \bar{f}_{\theta_k}$.

Given observation $y_{1:n}$, the log-likelihood function is

$$\ell_n(\vartheta_{0:n}) = \log \int \mu_{\vartheta_0}(x_0) \prod_{k=1}^n \bar{f}_{\vartheta_k}(x_k, y_k | x_{k-1}) d\vartheta_{0:n}. \quad (4)$$

Call $\nabla_i \ell_n(\vartheta_{0:n})$, the partial derivative of ℓ_n with respect to ϑ_i , $i = 0, \dots, n$. We write $\theta^{[k]}$ to denote k copies of θ . It is easy to check that

$$\nabla_\theta \ell_n(\theta^{[n+1]}) = \sum_{i=0}^n \nabla_i \ell_n(\vartheta_{0:n})|_{\theta^{[n+1]}}.$$

The goal is to approximate $\nabla_\theta \ell_n(\theta^{[n+1]})$. We construct the auxiliary model as follows. Let q be a mean-zero density on $\mathbb{R}^{d_\theta}$ with convex compact support and a positive definite covariance matrix Σ . For $\tau, \sigma > 0$, let $\kappa_{\theta_{0:n}, \tau, \sigma}$ be the density of the random vectors $\left\{ \theta_0 + Z_0, \theta_k + Z_0 + \sigma\tau^{-1} \sum_{\ell=1}^k Z_\ell, 1 \leq k \leq n \right\}$, where $Z_i \stackrel{i.i.d.}{\sim} q$. We introduce the auxiliary process $\{(\check{\Theta}_0, \check{X}_0), (\check{\Theta}_n, \check{X}_n, \check{Y}_n), n \geq 1\}$ as follows

- $\check{\Theta}_{0:n} \sim \kappa_{\theta_{0:n}, \tau, \sigma}$.
- The conditional density of $\{\check{X}_0, (\check{X}_k, \check{Y}_k), 1 \leq k \leq n\}$ given $\check{\Theta}_{0:n} = \vartheta_{0:n}$ is $\mu_{\vartheta_0}(x_0) \prod_{k=1}^n \bar{f}_{\vartheta_k}(x_k, y_k | x_{k-1})$.

We denote by $\check{\mathbb{E}}_{\theta_{0:n}, \tau, \sigma}$ the expectation operator with respect to this auxiliary process. When $\theta_i \equiv \theta$, which is the main case of interest, then we write $\check{\mathbb{E}}_{\theta, \tau, \sigma}$. Let us also define

$$\check{\Theta}_n^F(\theta) \stackrel{\text{def}}{=} \check{\mathbb{E}}_{\theta, \tau, \sigma}(\check{\Theta}_n | \check{Y}_{1:n}), \quad \text{and} \quad V_n^P(\theta) \stackrel{\text{def}}{=} \check{\mathbb{E}}_{\theta, \tau, \sigma} \left\{ (\check{\Theta}_n - \check{\mathbb{E}}_{\theta, \tau, \sigma}(\check{\Theta}_n | \check{Y}_{1:n-1}))^2 | \check{Y}_{1:n-1} \right\},$$

the filtering mean and the predictive variance of $\check{\Theta}_n$ in the auxiliary model when $\theta_{0:n} = \theta^{[n+1]}$ (with $\check{\Theta}_0^F(\theta) = \theta$). Precisely, notice that when $\theta_{0:n} = \theta^{[n+1]}$, $\check{\Theta}_{0:n}$ is then the random walk $\check{\Theta}_0 = \theta + \tau Z_0$, and for $1 \leq k \leq n$, $\check{\Theta}_k = \check{\Theta}_{k-1} + \sigma Z_k$. It is clear that this description falls in the framework developed above and we have the following result.

Theorem 2.1. *Suppose that for any $n \geq 1$, $\nabla \ell_n$ and $\nabla^2 \ell_n$ exists and are bounded on compact subset of $\{\mathbb{R}^{d_\theta}\}^{n+1}$. Suppose that $\sigma = \sigma(\tau) = o(\tau)$ as $\tau \rightarrow 0$. Let $\theta \in \mathbb{R}^{d_\theta}$. Then for any $n \geq 1$,*

$$\sum_{\ell=1}^n \{V_\ell^P(\theta)\}^{-1} (\check{\Theta}_\ell^F(\theta) - \check{\Theta}_{\ell-1}^F(\theta)) = \nabla_\theta \ell_n(\theta^{[n+1]}) + O(\tau + \sigma^2 \tau^{-2}), \quad (5)$$

as $\tau \rightarrow 0$.

Proof. For each $n \geq 1$, we apply Theorem 1.1 with $h(\vartheta_{0:n}) = \vartheta_n - \theta$ to get

$$\begin{aligned} \check{\Theta}_n^F - \theta &= \tau^2 \Sigma \left\{ \nabla_\theta \ell_n(\theta^{[n+1]}) + \left(\frac{\sigma}{\tau}\right)^2 \sum_{i=1}^n i \nabla_i \ell_n(\vartheta_{0:n})|_{\theta^{[n+1]}} \right\} + O(\tau^3) \\ &= \tau^2 \Sigma \left\{ \nabla_\theta \ell_n(\theta^{[n+1]}) + O(\sigma^2 \tau^{-2}) \right\} + O(\tau^3). \end{aligned} \quad (6)$$

Applied with $h(\vartheta_{0:n}) = (\vartheta_n - \theta)(\vartheta_n - \theta)'$, Theorem 1.1 yields

$$\check{\mathbb{E}}_{\theta, \tau, \sigma}((\check{\Theta}_n - \theta)(\check{\Theta}_n - \theta)' | \check{Y}_{1:n}) = \tau^2 \Sigma \left(1 + n \frac{\sigma^2}{\tau^2} \right) + O(\tau^3). \quad (7)$$

Now it is easy to check that

$$\begin{aligned} V_n^P(\theta) &= \sigma^2 \Sigma + \check{\mathbb{E}}_{\theta, \tau, \sigma}((\check{\Theta}_{n-1} - \theta)(\check{\Theta}_{n-1} - \theta)' | \check{Y}_{1:n-1}) \\ &\quad - (\check{\Theta}_{n-1}^F(\theta) - \theta)(\check{\Theta}_{n-1}^F(\theta) - \theta)'. \end{aligned}$$

The expansion in (6) implies that

$$(\check{\Theta}_{n-1}^F(\theta) - \theta)(\check{\Theta}_{n-1}^F(\theta) - \theta)' = O(\tau^4).$$

Combined with (7), this implies that

$$\tau^{-2} V_n^P(\theta) = \Sigma + \sigma^2 \tau^{-2} \Sigma + O(\tau) = \Sigma + O(\tau + \sigma^2 \tau^{-2}).$$

Since Σ is non-singular, for τ small enough, $\tau^{-2} V_n^P(\theta)$ is also non-singular and

$$\{\tau^{-2} V_n^P(\theta)\}^{-1} = \Sigma^{-1} + O(\|\tau^{-2} V_n^P(\theta) - \Sigma\|) = \Sigma^{-1} + O(\tau + \sigma^2 \tau^{-2}).$$

It follows that

$$\begin{aligned} \sum_{\ell=1}^n \{V_\ell^P(\theta)\}^{-1} (\check{\Theta}_\ell^F(\theta) - \check{\Theta}_{\ell-1}^F(\theta)) &= \sum_{\ell=1}^n \tau^{-2} \Sigma^{-1} (\check{\Theta}_\ell^F(\theta) - \check{\Theta}_{\ell-1}^F(\theta)) \\ &\quad + \tau^{-2} \sum_{\ell=1}^n \left(\{\tau^{-2} V_\ell^P(\theta)\}^{-1} - \Sigma^{-1} \right) (\check{\Theta}_\ell^F(\theta) - \check{\Theta}_{\ell-1}^F(\theta)). \end{aligned}$$

From (6), we know that $\check{\Theta}_\ell^F(\theta) - \check{\Theta}_{\ell-1}^F(\theta) = O(\tau^2)$ from which we deduce that

$$\tau^{-2} \sum_{\ell=1}^n \left(\{\tau^{-2} V_\ell^P(\theta)\}^{-1} - \Sigma^{-1} \right) (\check{\Theta}_\ell^F(\theta) - \check{\Theta}_{\ell-1}^F(\theta)) = O(\tau + \sigma^2 \tau^{-2}).$$

Again, using (6) and by telescoping the sum, we write

$$\sum_{\ell=1}^n \tau^{-2} \Sigma^{-1} (\check{\Theta}_\ell^F(\theta) - \check{\Theta}_{\ell-1}^F(\theta)) = \tau^{-2} \Sigma^{-1} (\check{\Theta}_n^F - \theta) = \nabla_{\theta} \ell_n(\theta^{[n+1]}) + O(\tau + \sigma^2 \tau^{-2}).$$

The theorem is proved. □

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